

CompSci 101

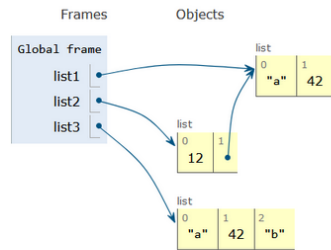
Introduction

CompSci 101, Spring 2025

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```
1 list1 = ['a', 42]
2 list2 = [12, list1]
→ 3 list3 = list1 + ['b']
```



CompSci 101: Introduction to Computer Science

Course Description

Introduction to practices and principles of computer science and programming and their impact on and potential to change the world. Algorithmic, problem-solving, and programming techniques in domains such as art, data visualization, mathematics, natural and social sciences.

Programming using high-level languages and design techniques emphasizing abstraction, encapsulation, and problem decomposition. Design, implementation, testing, and analysis of algorithms and programs. No previous programming experience required. For this version of the course, you will learn the programming language Python.



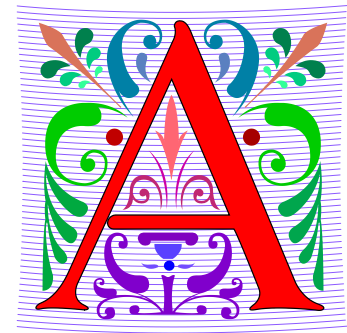
Susan Rodger
January 9, 2025

Every lecture:

Sit in the
front 6 rows
(the room is way too big!)

Learn the CS Alphabet

A is for ...



- **Algorithm**
 - Step-by-step instructions realized in a program
- **Abstraction**
 - Hiding things is powerful
 - “What” vs “How”
- **APT**
 - Algorithmic Problem-solving Testing
- **API**
 - Application Programming Interface - using Libraries

Feature someone
related to CS in every
lecture

Plan for Today!

- Introduce course staff
- Course logistics
- What is computer science about?
- Introduce class activities
- Know what work you should be doing

Introduce Course Staff

About Prof Rodger



Course Staff – Teaching Associate

- Violette Walker
- Handles many logistics!
 - SDAO Accommodations
 - Hiring of UTAs
 - Organizing staff



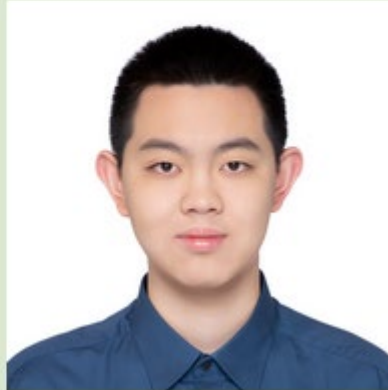
More Course Staff

- Graduate TAs

**Graduate TA:
Weihang Guo**



**Graduate TA:
Jiasheng Wang**



And More Course Staff

- About 20 Undergraduate Teaching Assistants

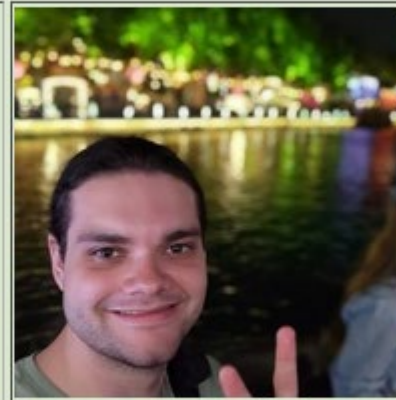
HEAD Undergraduate TAs (UTAs)



Kevin Alvarenga
Labs



Janet Jiang
Consulting Hours



Nolan Potter
Grading

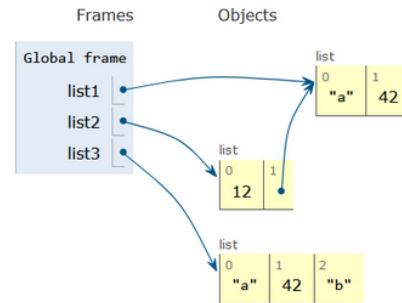
Course Logistics

Go over CompSci 101 webpages

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Announcements

- Check out the calendar on the course website
 - PRE-WORK – what you must do before the lecture
 - LECT/LAB – will put notes/videos here from the live lecture or for the lab
 - DUE – what is due each week.
- What has been updated?
- Assignment 1 is already out!
- PreLab 1 due tonight (or 8am Fri)
- Prelab for Lab2 (install Python)
- You will see a link to this video!

CompSci 101, Spring 2025
CompSci 101 Calendar

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All materials we use in class are accessible via this page.
This is a temporary schedule for what we will do, subject to change!
[Information about grading and course logistics.](#)

	January 6-10 Week				
	Monday	Tuesday	Wednesday	Thursday	Friday
PRE-WORK	1/6	1/7	1/8	1/9 Topics: Introduction to the course Read Course Webpages QZ01 on Canvas	1/10
LECTURE LAB				First Lecture	Prelab for Lab 1 (do before Friday!) (by 8am Friday ok) Lab 1
ASSIGNMENTS APTS DUE				Assign 1 out	

	January 13-17 Week				
	Monday	Tuesday	Wednesday	Thursday	Friday
PRE-WORK	1/13	1/14 Topics: Python, Variables, Operators, String operations Textbook • 1.1-1.5 • 2.1-2.7 • 2.9-2.11 • 9.3	1/15	1/16 Topics: Functions, Parameters, Scope, Function Composition Textbook/Document • Functions Document • 6.2 • 6.4	1/17

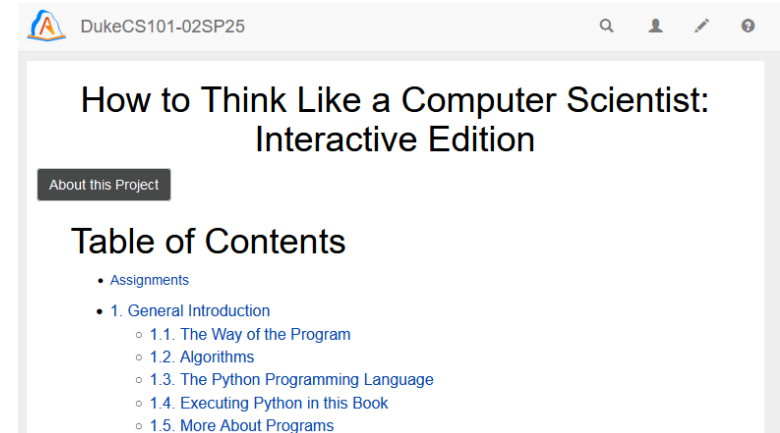
Course overview, logistics

www.cs.duke.edu/courses/spring25/compsci101

- **Programming assignments: APTs and Assignments**
 - Acknowledge assistance, to learn to program ...
 - Be aware of late policy
- **Labs**
 - Attend each Friday
- **Lecture - Classwork**
 - Attend the live lecture - participate
 - If you can't attend you must watch it and participate within 24 hours
- **Exams: 3 exams and final**
 - All old exams available

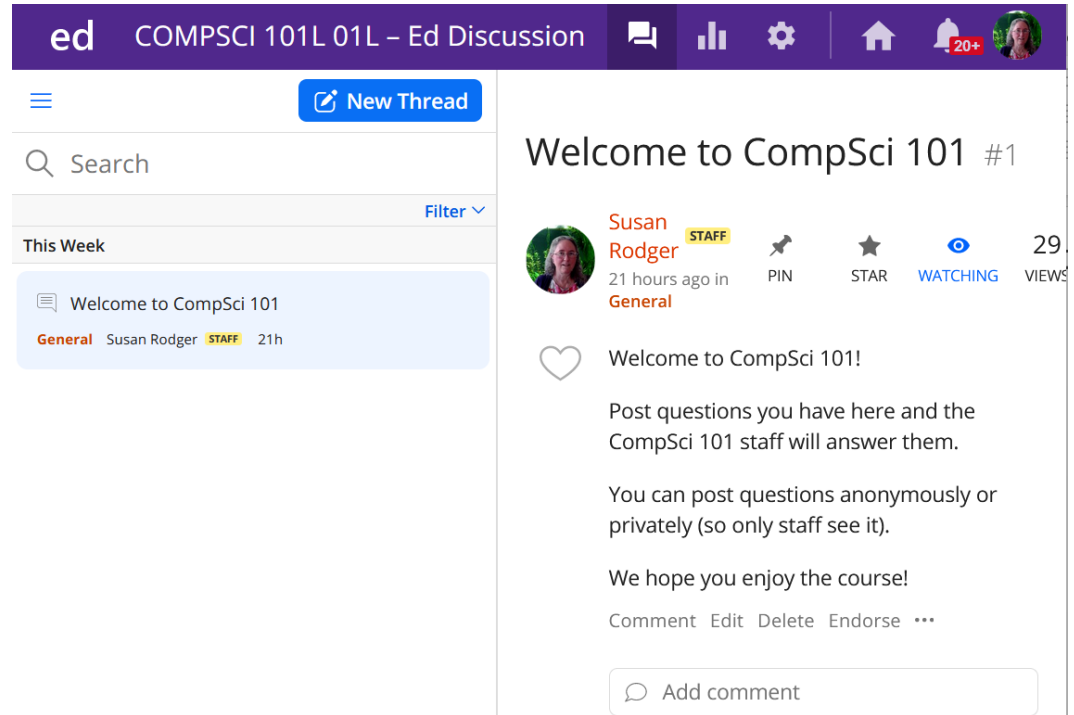
Course Textbook in Runestone

- Your account already created!
- You got email from Joe Shamblin
- See instructions in Canvas on how to login
- Use book for
 - Reading before quizzes
 - In-class activities



Questions?

- Don't send us email!!!!!!!!!!!!!!
- Post your questions on Ed Discussion
 - We will answer them there!
 - You should try to answer them too
 - Want to be a UTA one day? Answer questions!
- Post Questions during lecture!



What's in Compsci 101?

- Learning about computing, computer science, and programming
 - Vocabulary of Python and programming
 - *Power of automation, repetition, scale*
 - Understanding and changing the world
- **Programming using Python**
 - Tools: PyCharm, Libraries, ...
 - Using mathematical and scientific techniques
 - *Art and science* of programming

What is Computer Science about?

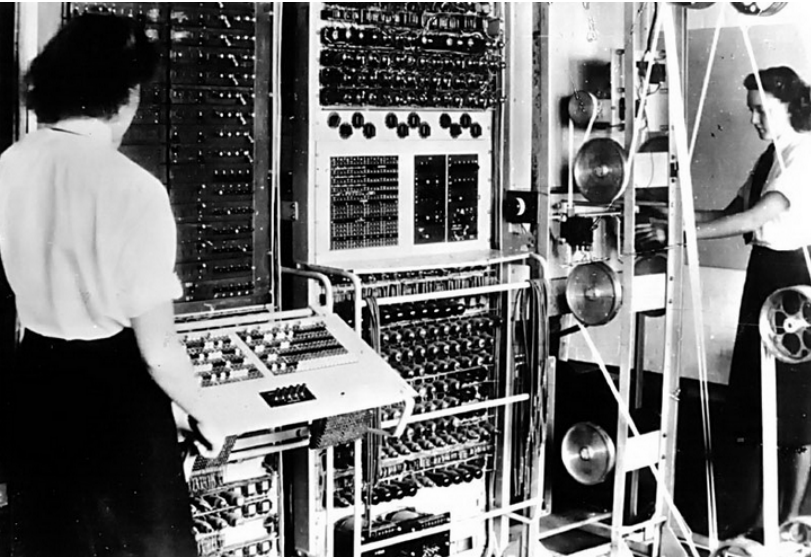
WOTO – Working Together

<http://bit.ly/101s25-0109-1>

Discuss with others, then everyone fills out their own form.

What is Computer Science?

How it started How it's going



Computers speak in 0's and 1's

- In old computers
 - Control electric current using the vacuum
- Nowadays, use switches
 - A switch that is "on" or "closed" represents
 - Passes electrical current through
 - A switch that is "off" or "open" represents 0
 - Blocks electrical current
 - Express 0's and 1's, called bits
 - 8 bits are a byte and represent a symbol
- What letter is 01000001 ?

1906



What is Computer Science now?

- Artificial Intelligence



Perseverance
Mars Rover

Self-driving
car



Roomba



Personal Robot

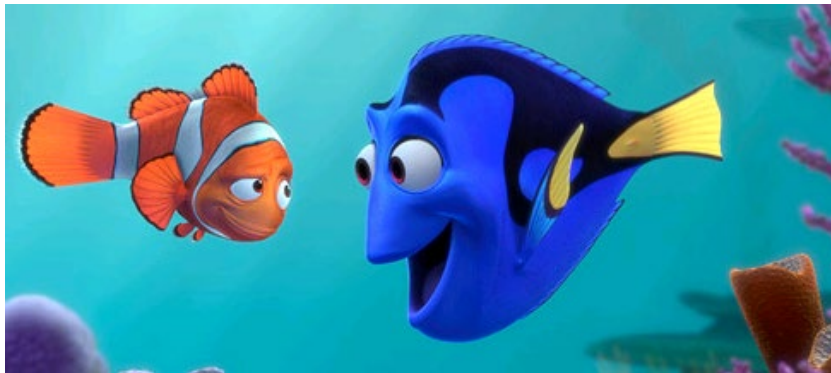
What is Computer Science?

- Medicine, Genomics



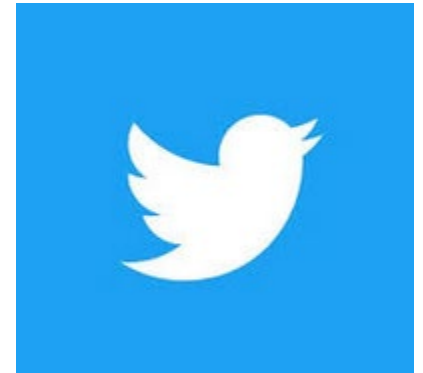
What is Computer Science?

- Animation



What is Computer Science?

- The Organization of Data, Sharing, and Searching



Prerequisites for Compsci 101



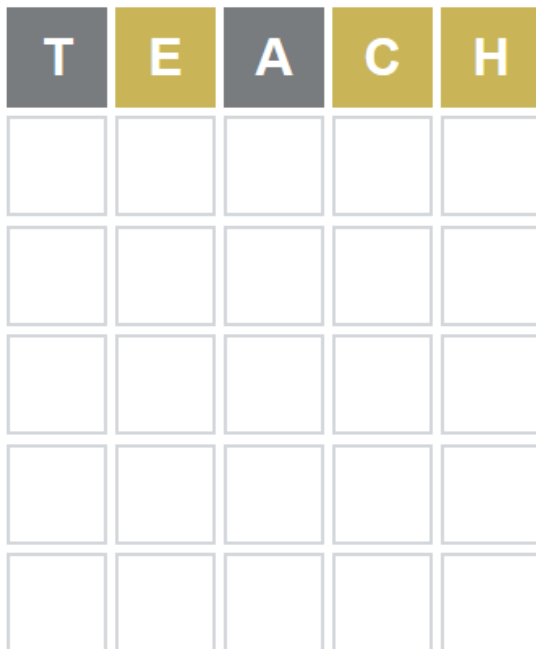
After taking this course you will be able to

- Write a program for Wordle
- Write a word finder to help someone solve Wordle

Wordle

Q	W	E	R	T	Y	U	I	O	P
A	S	D	F	G	H	J	K	L	
ENTER	Z	X	C	V	B	N	M	⌫	

Wordle



Wordle Solver

This simple to use tool will help you solve any Wordle answer you are having trouble with. It will enter in the letters that you have gotten in the right spot into the same spot in the correct Letters section. Add any letters that are in the puzzle but not in the right spot to the Misplaced Letters area. Finally, enter anything you've guessed that isn't right at all into the Incorrect Letters section. The potential answers will populate as you enter in letters, eventually giving you the solution!

Correct Letters



Misplaced Letters



Incorrect Letters



Potential Answers (10)

cheek cheer chess chide chief chime choke
chore chose niche

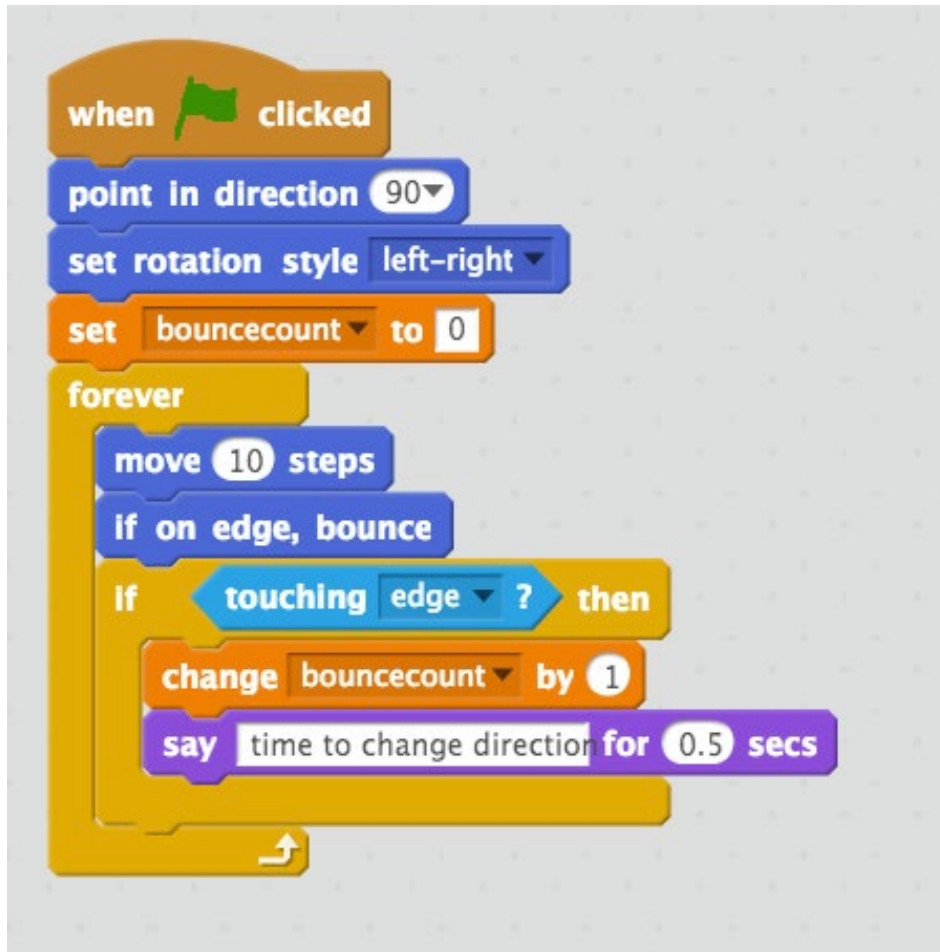
Recommended Guesses

CHIDE CHIEF CHIME NICHE

Common Letters

E (12) C (10) H (10) I (4) S (3) O (3) K (2)
R (2) M (1) F (1)

What does this program do?



- "Hello World"
- Scratch Program
- Colors
 - Duke blue: motion
 - Mustard: control
 - Light blue: sensing
 - Orange: data
 - Purple: looks

WOTO: WOrking TOgether

<http://bit.ly/101s25-0109-2>

Analyze this
Scratch Program?

What language will we learn?

- <http://www.python.org/>
- Python is a *multi-paradigm* language
 - Procedural
 - Functional
 - Object-Oriented
- Simple, libraries, widely used
- Guido van Rossum



Python code

hello.py

```
1  """
2      Created on 1/12/2023
3
4      @author: Susan H. Rodger
5  """
6
7  ▶ if __name__ == '__main__':
8      print("Hello CompSci 101 students!")
```

OUTPUT:

Python Code, second program

```
6  def greeting(name):  
7      print("Hello " + name)  
8  
9  if __name__ == '__main__':  
10      greeting("CompSci 101!")  
11      greeting("Beenie, Keeah and Moe")
```

OUTPUT:

Your Tasks!

- **Login to the textbook - start reading!**
 - Your account emailed, Canvas announcement
- **PreLab 1 Warm-up**
 - Due tonight (really by 8am tomorrow)
- **Go to Lab 1 on Friday**
- **Lecture 2 – Reading and Quiz**
 - Do by 10am Tuesday, Jan 14
- **Start Assignment 1 – Blockly**
 - Due 1/16 at 11:59pm
- **Install Pycharm**
 - See Pre-Lab 2